

# DMS DEVELOPMENT

## Documentation


### Abstract

In this module I learn about DMS project structure and database structure



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# DATABASE STRUCTURE

## 1.1 TYPES OF DATABASE

- We have three types of database.
  - Company level
  - System level
  - Application level

**Company level:**

- It's contains two types of tables,
  - Company 's tables
  - Year's tables
- It format is: dms_companyId_financialyr
- Ex: dms_000197_2324, dms_000198_2324, dms_000197_2425,

**System level:**

- It contains system tables.
- It contains default data, which is shared by all clients.
- It is named as a dmssys in database.

**Application level:**

- It contains two type of table,
  - User detail tables
  - Company detail tables
- It contains relationships between user and company tables.
- In DMS we marge this level with System level.
- Ex:
  - SMH01, SMH02, SMH03…

## 1.2 TABLE NAMING CONVENTION

- We take first character based on table type,
  - If Company then, C
  - If Year then, Y
  - If System then, S
- Then we take second character based on grouping of table,

- If Master then, M
- If Transaction then, T
- If Setup then, S

- After we take third character based in sub grouping of table,
- After we take two character for sequence.

**Example:**

- YTT41
  - Y means Year level table
  - T means transaction group
  - T means Transaction Main sub group
  - 41 is sequence for table
- YMA01:
  - Y mean Year level table
  - M means Master group
  - A means Account Master sub group
  - 01 is sequence
- CMM01:
  - C means Company level table
  - M means Master group
  - M means General Master sub group
  - 01 is sequence
- We can't take last three character as duplicate across the all databases.

## 1.3 Table Naming Convention

- We have three types of fields in table,
  - Normal Field
  - Common Field
  - Unique DMS Field

**Normal Field:**

- First, we take last three character of table name.
- Then we put F character for Filed.
- After we take two characters for sequence

**Example:**

- If we have table YMM41, then fields are,
    - M41F01, M41F02, M41F03…

**Common Field:**

- We have some common fields which is used in most of table so we should use separate naming convention for those fields,
- First three character same as normal field
- After three character we should use based on this table

| Field Name | Character |
|---|---|
| **DealerId** | C01 |
| **SalesmanId** | C02 |
| **VoucherType** | C04 |
| **SubVoucherType** | C05 |
| **UserId(Creation)** | C06 |
| **DateTime(Creation)** | C07 |
| **UserId(Modification)** | C08 |
| **DateTime(Modification)** | C09 |

**DMS Unique Field:**

- We have some unique field which is not in cloud Miracle but it is exist in DMS.
- First three character same as normal field
- After we take X as third character
- Then we take two characters for sequence.

**Example:**

- If we have YTT41 table and we want to create name for those fields which is not in Cloud Miracle but exist in DMS then,
- Name becomes T41X01

## 1.4 User field and Expanse detail table

**Table:**

- We normally create new table for User field / Expanse for any existing master / transaction group of table.
- Naming of that table will be: Main Table name + “U/E”

**Example:**

- User field table for account master will be YMA01U whose main table is YMA01
- Expense table for item wise expense will be YTT02E whose main table is YTT02

**Field:**

- First field we should take last three character of table name
- After we use “U/E” respectively for User field and Expense.
- Then we take two characters for sequence.

**Example:**

- For User field YTT02 is T02U01.
- For User field YMA01 is A01U01.

## 1.5 Pseudo Name

- For security purpose we should use pseudo name in GUI.
- First we take first three character as it is.
- Then third is character is base on type of field,
  - If common field C, then 0
  - If normal field F, then 1
  - If DMS unique field X, then 2
- Then we use two character for sequence.

**Example:**

- T41C01 then, T41001
- T41F01 then, T41101
- T41X01 then, T41201

## 1.6 Datatype Guidelines

- If master should have less than 65,000 record then we use SMALLINT instead of INT in Primary Key.
- If any field would be contains two character or less than two character then we must use Char(1) and Char(2).
- Null value not allow in any field, we must try to set appropriate default value if possible.
- Don’t add comment into field.

## 1.7 MySQL Guideline

### 1.7.1 Stored Procedure:

- Parameter name start with small ‘p’ with underscore (_)
- Ex: p_B01F01, p_B02F02
- Use Proper Datatype for Parameter
- Variable name start with small ‘v’ with underscore (_)
- Ex. v_UserId;
- Use Default Value if possible
- Ex. v_UserId INT DEFAULT 0;
- Don’t Used and avoid @Session Variable use local variable.
- Drop temporary table after usage are complete.

### 1.7.2 Select Statement:

- Don’t Used and avoid COUNT (*) used COUNT with Key Column.

**Format:**

```
SELECT
        Fields
FROM
        Table Name (Alias)
        JOIN
WHERE
        Condition
Group BY
        Fields
Order By
        Fields
```

**Where Clause**

- Used Primary key or Index Column First
- Date Rang using between must in bracket
- Apply filter with proper sequence , like,
    - Dealer Id
    - Salesman Id
    - Date
    - Sub voucher type

**Join:**

- Used Column they belong Primary Key or Index

- Used Column with same data type
- If we should do multiple join on same table then, we must give appropriate alias.
- Ex:
  - Left Join YMP16 P16A on P16A. P16F01 = P01F020
  - Left Join YMP16 P16B on P16B. P16F01 = P01F024

**Order By Clause**

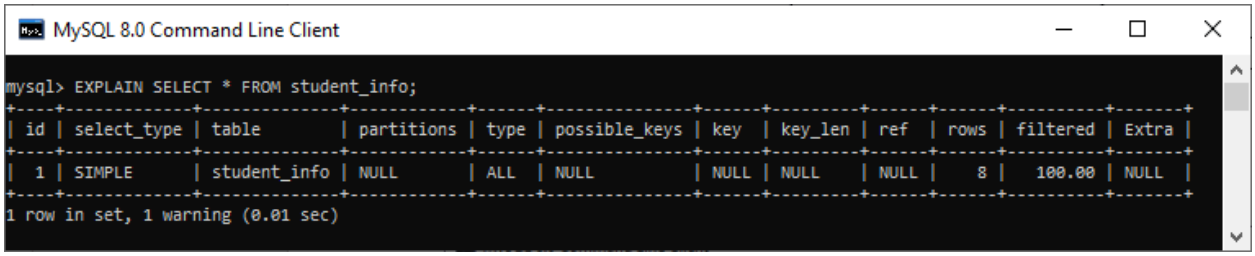
- Used only necessary
- Used only required column

**Unique Row**

- Must use Limit 1 if we want to get single record in API.

**Explain with SELECT**

- The EXPLAIN keyword is used to obtain information about how our SQL databases execute the queries in MySQL.




### 1.7.3 Temporary Table:

- Used when required.
- Always take required columns.
- Use primary key or index for batter performance.
- Don't restructured in between operation, it's impact on performance.
- Drop after usage.

### 1.7.4 Other guideline

**Data Duplication**

- Column Verification and avoid same data column

**Column Alias**

- Used Column Alias only necessary

**Function with WHERE**

- Used only necessary

**Find_In_Set()**

- Don't Used and avoid find_in_set() functions.

## 1.8 IMPORTANT TABLE

| Table Name | Original Name |
|---|---|
| **SMH01** | Company Details |
| **SMH02** | User Details |
| **SMH03** | Year Details |
| **SMZ01** | Dropdown Master |
| **SMZ02** | Company Master |
| **SMZ03** | State Master |
| **SMZ09** | Company / Year Database structure changes details |
| **SMZ11** | Database Table Definition Detail |
| **SMZ12** | Database Table Field Definition Detail |
| **SMZ13** | Parent Child Master |
| **SMZ14** | Account Group Master |
| **SMZ24** | Activity Master |
| **SMZ25** | Manu Master |
| **SMZ26** | User Group Master |
| **SMZ27** | User Master |
| **SMZ28** | User Login Details |
| **SMX01** | Setup Configuration |
| **SMX02** | Setup Main Category Detail |
| **SRR01** | Template Field Master |
| **CMB09** | Activity Master |
| **CMB10** | Manu Master |
| **CMU01** | User Group Master |
| **CMU02** | User Master |
| **YMA03** | Salesman Master |
| **YMA04** | Route Master |
| **YMA05** | Dealer – Salesman Mapping |
| **YMA06** | Party Master |

# DMS Development

| | |
|---|---|
| **YMA07** | Country Master |
| **YMA08** | State Master |
| **YMA09** | District Master |
| **YMA10** | Taluka Master |
| **YMA11** | City Master |
| **YMA12** | Zone Master |
| **YMA13** | Region Master |
| **YMA14** | Area Master |
| **YLL01** | User Master Log |